Kafka Learning Path for a Senior Platform Engineer

Topics per course — focused on cluster administration, provisioning, reliability, monitoring and operations; application development is intentionally minimized.

Course	Recommendation	Target time
1. Learn Apache Kafka for Beginners v3	Selective — foundations, CLI, admin/config concepts	3.5–4 h
2. Kafka Cluster Setup & Administration	Core — almost all sections	3–3.5 h
3. Kafka Monitoring & Operations	Core — complete	4–4.5 h
4. Kafka Connect Hands-on Learning	Selective — if platform owns Connect	2.5–3 h
5. KSQL on ksqlDB	Skip completely	0 h
6. Schema Registry & REST Proxy	Selective — infrastructure concepts	1.5–2 h

1. Learn Apache Kafka for Beginners v3

Verdict: SELECTIVE

Kafka Introduction / Fundamentals / Theory

• Kafka architecture: broker, topic, partition, replica, leader/follower

• Offsets, consumer groups, replication factor and message ordering

• Retention, durability and basic Kafka internals

Starting Kafka

• Broker startup/shutdown and basic environment setup

• Basic broker configuration and cluster interaction

CLI 101 — MUST WATCH

• Topic creation, listing, describing and altering topics

• Producing/consuming messages from CLI

• Consumer groups, offsets, partition and broker metadata

Kafka UI / Conduktor

• Basic cluster visibility and operational inspection

Kafka in the Enterprise for Admins

• Administrative considerations, sizing and production concepts

Advanced Topic Configurations

• Retention and cleanup policies: delete vs compact

• Segment concepts and retention limits

• min.insync.replicas and unclean leader election

Producer / Consumer configuration concepts — DO NOT learn coding

• Producer: acks, retries, idempotence, compression, batch.size, linger.ms

• Consumer: group.id, auto.offset.reset, enable.auto.commit

• Consumer polling, heartbeat/session settings, rebalancing and lag

SKIP

- Kafka Java Programming 101
- Deep producer/consumer coding
- Kafka Extended APIs for Developers
- Deep application implementation in the Wikimedia/OpenSearch examples

2. Kafka Cluster Setup & Administration

Verdict: CORE — ALMOST COMPLETE

Course Introduction & Architecture

- Multi-broker architecture, cluster components and communication
- Partitions, replication, leaders/followers and cluster topology

ZooKeeper Quorum Setup — LEGACY KNOWLEDGE

- ZooKeeper quorum and Kafka coordination
- Understand historical ZooKeeper-based Kafka; prioritize KRaft knowledge separately

Kafka Cluster Setup — MUST WATCH

- Broker provisioning and configuration
- Listeners and advertised listeners
- Inter-broker communication
- Storage/log directories
- Partitions, replicas and replication
- Cluster formation, startup/shutdown and operational validation

Operational focus

- Broker configuration and cluster health
- Network and storage considerations
- Production deployment considerations

SKIP / LOW PRIORITY

- Code download and repetitive setup demonstrations
- Next-steps promotional material

3. Kafka Monitoring & Operations

Verdict: CORE — COMPLETE

Kafka Quick Setup in AWS

• AWS deployment considerations

• Networking, storage and broker placement

• Connectivity and infrastructure considerations

Kafka Administration Setup

• Broker/topic administration

• Cluster configuration and operational commands

Kafka Monitoring — Prometheus + Grafana — HIGHEST PRIORITY

• Broker, JVM, topic, partition and consumer metrics

• CPU, memory, disk and network monitoring

• Request rate/latency and Kafka performance metrics

• Under-replicated partitions, offline partitions and ISR health

• Leader distribution and controller-related metrics

• Consumer lag monitoring and alerting

• Prometheus/Grafana dashboards and operational alerting

Kafka Operations — HIGHEST PRIORITY

• Broker failure and recovery

• Adding/removing/replacing brokers

• Partition reassignment and replica movement

• Leader movement, ISR problems and cluster degradation

• Disk/storage problems and operational troubleshooting

• Consumer lag and production incident investigation

Kafka Cluster Upgrade — MUST WATCH

• Rolling upgrades and compatibility

• Broker-by-broker upgrade strategy

• Validation, downtime avoidance and rollback considerations

SKIP

• Only repetitive installation/code-download material and next-steps material

4. Kafka Connect Hands-on Learning

Verdict: SELECTIVE — IF PLATFORM OWNS CONNECT

Kafka Connect Concepts — MUST WATCH

• Kafka Connect architecture

• Worker, connector and task

• Source vs sink connectors

• Standalone vs distributed mode

• Connect REST API

Connect Cluster Setup — MUST WATCH

• Distributed Connect cluster

• Worker configuration and scaling

• Internal topics: connect-configs, connect-offsets, connect-status

Troubleshooting Kafka Connect — MUST WATCH

• Worker failures

• Connector/task failures

• Task rebalancing and restarts

• Offset and connectivity issues

• Plugin/configuration problems

Source / Sink Hands-on — SELECTIVE

• Understand operational behavior and failure modes

• Skip application-specific implementation depth

SKIP

• Writing your own Kafka Connector

• Custom Java connector development

• Deep source/sink application coding

5. KSQL on ksqlDB for Stream Processing

Verdict: SKIP COMPLETELY

Why skip

• Focused on SQL-based stream processing and application/business logic

• Not a priority for a Platform Engineer focused on Kafka infrastructure and operations

No required topics

• Save the full course time for cluster operations, monitoring and troubleshooting

6. Confluent Schema Registry & REST Proxy

Verdict: SELECTIVE — INFRASTRUCTURE ONLY

Schema fundamentals

• Understand what schemas are and why Kafka platforms use Schema Registry

• Avro fundamentals at a conceptual level

Schema Registry + Kafka — MUST WATCH

• Schema Registry architecture and Kafka integration

• _schemas topic concept

• High availability and operational architecture

• Schema IDs and producer/consumer interaction

Compatibility & Schema Evolution — MUST KNOW

• Backward, Forward, Full and None compatibility

• Schema evolution and deployment impact

• Troubleshooting serialization/deserialization failures

REST Proxy — BASIC

• Purpose and architecture

• Operational deployment and HTTP access to Kafka

SKIP

• Avro in Java

• Java object implementation

• Deep application development

Senior Platform Engineer — Priority Order

1. Kafka Monitoring & Operations — highest priority
2. Kafka Cluster Setup & Administration
3. Kafka Fundamentals + CLI
4. Kafka Connect operations — when owned by the platform team
5. Schema Registry / REST Proxy operations
6. ksqlDB — skip

Production troubleshooting outcomes to target

- Consumer lag increasing continuously
- Under-replicated partitions / ISR reduction
- Broker failure and broker replacement
- Uneven disk usage across brokers
- Kafka disk approaching capacity
- Producer NotEnoughReplicasException
- Consumer group continuously rebalancing
- Partition reassignment and broker removal
- Rolling Kafka version upgrade without downtime
- Kafka latency increase: isolate broker, network, producer and consumer causes
- ZooKeeper vs KRaft architecture and migration concepts
- Schema compatibility or deserialization failure after an application deployment

Target study time: approximately 14–17 hours instead of completing all six courses (~30.5 hours).

Learning objective: operate, monitor, troubleshoot, scale, upgrade and recover a production Kafka platform — without spending time becoming a Kafka application developer.